(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendruthy (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.KONDETI GANESH**

Registration Number: **323129520074**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies (CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

SMART WATER TANK LEVEL MONITORING AND AUTOMATIC PUMP CONTROL SYSTEM TO PREVENT OVERFLOW AND WATER SHORTAGE

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mrs.Chinnammulu

Department of MECHANICAL ENGINEERING

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr.KONDETI GANESH

Reg.No: 323129520074

Department of MECHANICAL ENGINEERING

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Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compul-sory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or indus- try.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and com-municate regularly with your supervisor to provide updates on your progress with work assignments.

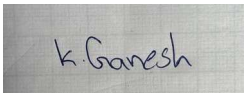
Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever nMECHANICALssary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organiza- tion.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organi- zation.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

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Student's Declaration

I, **Mr.KONDETI GANESH**, a student of **Bachelor of Technology** Program, Reg. No. **323129520074** of the Department of **MECHANICAL ENGINEERING** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mrs.Chinnammulu**, Department of **MECHANICAL ENGINEERING**, **Wellfare Institute of Science, Technology and Management**.



(Signature and Date)

Official Certification

This is to certify that **Mr.KONDETI GANESH** , Reg. No. **323129520074** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **SMART WATER TANK LEVEL MONITORING AND AUTOMATIC PUMP CONTROL SYSTEM TO PREVENT OVERFLOW AND WATER SHORTAGE** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **MECHANICAL ENGINEERING** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsements

Faculty Guide

Head of the Department

Head of the Department
Department of Mechanical Engineering
Wellfare Institute of Science
Technology and Management
Pinagadi, Visakhapatnam
Andhra Pradesh-531173

Principal

Certificate from Intern Organization

This is to certify that **Mr.KONDETI GANESH**, Reg. No. **323129520074** of **Wellfare Institute of Science, Technology and Management**, underwent internship in **SMART WATER TANK LEVEL MONITORING AND AUTOMATIC PUMP CONTROL SYSTEM TO PREVENT OVERFLOW AND WATER**

SHORTAGE And Sentiment Analysis For Social Media Applications at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satis-factory** (Satisfactory/~~Not Satisfactory~~).



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Authorized Signatory with Date and Seal

Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mr.R.Sagar**, Head of the Department of **MECHANICAL ENGINEERING**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mrs.Chinnammulu**, Assistant Professor of the Department of **MECHANICAL ENGINEERING** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my 8-week Short-Term Internship in developing a Smart Water Tank Level Monitoring and Automatic Pump Control System. The internship was undertaken as part of the academic curriculum. The primary objective of this internship was to gain proficiency in Internet of Things (IoT), data analytics, and automated control systems to address the critical issue of water resource management.

Efficient water management is essential for residential, commercial, and industrial facilities to prevent water wastage and ensure a continuous water supply. Traditional water tank management relies on manual monitoring and manual pump operation, which often leads to tank overflows, water shortages, and unnecessary electricity consumption. This project successfully addresses these limitations by providing an intelligent system that continuously monitors water levels and automatically controls pump operation based on predefined thresholds.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

• To design and implement a water management simulation using Python that models real-world tank capacities, inflow/outflow rates, and pump dynamics.

• To integrate simulated IoT sensors (ultrasonic, pressure, float switch) for comprehensive water level monitoring and fault detection.

• To develop algorithms for automatic pump control that prevent both dry running (water shortages) and overflows.

• To create interactive data visualizations using Matplotlib and Seaborn to display water level trends, pump operation cycles, and system efficiency.

• To understand the software development lifecycle from requirements gathering to testing and performance evaluation.

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1.2 Outcomes Achieved

Key outcomes from my internship include:

• A fully operational simulation of a Smart Water Tank Management System capable of managing residential, commercial, industrial, and institutional tanks simultaneously.

• An advanced analytics module that generates insights on daily water usage patterns, pump runtimes, and alert distributions.

• Successful implementation of an automated control algorithm that maintains water levels within safe operational boundaries (typically between 25% and 90% capacity).

• Professional dashboards and visualizations that help facility managers optimize water resources and electricity usage.

• Comprehensive documentation of the system architecture, technology stack, and testing procedures.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The organization where this internship was conducted focuses on bridging the academia-industry divide, enhancing student employability, promoting innovation, and fostering an entrepreneurial ecosystem. By leveraging emerging technologies such as IoT and AI, the organization aims to augment and upgrade the knowledge ecosystem, enabling beneficiaries to become contributors themselves. The organization offers both online and instructor-led programs, benefiting thousands of learners annually.

2.2 Vision, Mission, and Values

• Vision: To combine cutting-edge technology with impactful social ventures to drive technological prosperity and sustainable resource management.
• Mission: To support individuals dedicated to helping others by empowering and equipping them with modern technical skills, thereby creating an extensive educational network dedicated to societal betterment and environmental conservation.
• Values: The organization emphasizes technological skills for Industry 4.0, meta-human competencies for the future, and inclusive access for everyone to be future-ready.

2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the mission. Interns are expected to adhere to the following policies:
• Confidentiality: Interns must maintain the confidentiality of all organizational data and sensitive information.
• Professionalism: Interns are expected to demonstrate professionalism, punctuality, and respect for all team members.
• Learning and Contribution: Interns are encouraged to actively participate in projects, share ideas, and contribute to the organization's goals.

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2.4 Organizational Structure

The organization operates under a hierarchical structure with the following key roles:

- Board of Directors: Provides strategic direction and oversight.
- Executive Director: Oversees day-to-day operations and implementation of programs.
- Program Managers: Lead specific initiatives and technology projects.
- Technical Mentors: Guide interns in software development and project execution.
- Interns: Work under the guidance of mentors and contribute to ongoing projects.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are typically placed under the guidance of technical mentors or project managers. The roles and responsibilities of these guiding employees include:

1. Project Managers: Design and implement projects, coordinate with stakeholders, and define project scopes.
2. Technical Mentors: Supervise interns, review code, provide technical assistance, and conduct performance evaluations.
3. Research Analysts: Provide domain knowledge, in this case, regarding fluid dynamics, pump specifications, and IoT standards.

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CHAPTER 3

PROBLEM ASSESSMENT

3.1 Problem Statement

Efficient water management is essential for residential, commercial, and industrial facilities to prevent water wastage and ensure a continuous water supply. Traditional water tank management relies on manual monitoring and manual pump operation, which often leads to tank overflows, water shortages, and unnecessary electricity consumption. Building owners and facility managers require intelligent systems that can automatically monitor water levels and control water pumps in real time.

The proposed solution is a Smart Water Tank Level Monitoring and Automatic Pump Control System that continuously monitors water levels and automatically controls pump operation using IoT technology. The system provides a centralized platform where water tank status is monitored through a secure and user-friendly interface.

3.2 Key Parameters (PC1)

In accordance with the evaluation criteria (PC1), the key parameters of the problem have been identified as follows:

Issue to be Solved: The reliance on manual pump operation which results in human error —either forgetting to turn the pump on (causing water shortages) or forgetting to turn it off (causing overflows and water wastage).

Target Community: The target community includes homeowners, apartment complex associations, industrial plant managers, hospital administrators, and educational institution facility teams.

User Needs and Preferences:

- Facility managers need automated control to eliminate manual intervention.
- Property owners require real-time visibility into water consumption patterns.

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• Maintenance teams need instant alerts for pump malfunctions or sensor faults.
• All users prefer a system that reduces electricity bills by optimizing pump runtimes.

Environmental Impact: Tank overflows contribute significantly to fresh water wastage. By automating pump shut-offs and tracking usage patterns, the system directly supports water conservation and sustainable resource management.

3.3 Requirements Evaluation (PC2)

In accordance with the evaluation criteria (PC2), the requirements have been evaluated and mapped to the problem statement:

Functional Requirements:

1. Real-time Monitoring: The system must continuously measure water levels across various tank types (residential, commercial, industrial).
2. Automated Control: The system must automatically switch pumps ON at low thresholds (e.g., 25%) and OFF at high thresholds (e.g., 90%).
3. Anomaly Detection: The system must automatically detect pump malfunctions (pump ON but level dropping), sensor faults, and battery issues.
4. Data Visualization: The system must provide interactive dashboards displaying water levels, pump status, usage history, and alerts.

Non-Functional Requirements:

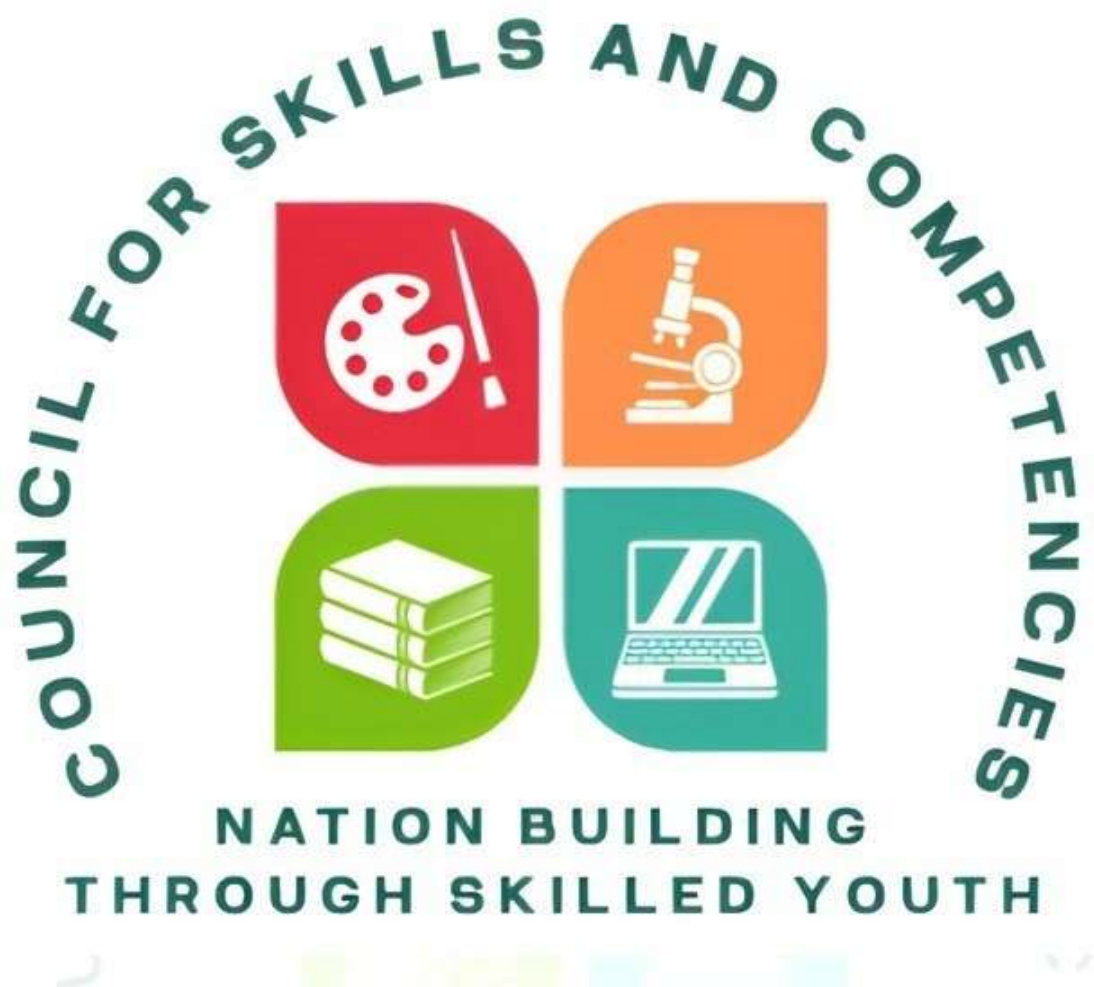
1. Scalability: The architecture must support monitoring multiple tanks across large commercial or industrial complexes.
2. Reliability: The automated control logic must be fail-safe to prevent dry running of pumps.
3. Accuracy: Water level sensors must be highly accurate to prevent premature or delayed pump switching.
4. Security: Pump control commands must be securely transmitted to prevent unauthorized manipulation.

Mapping to Problem Statement:

The functional requirements directly address the "manual monitoring and manual pump

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operation" by providing automated, threshold-based control. The non-functional requirements ensure the system is "secure, scalable, and suitable for homes, apartments, industries" as outlined in the problem statement.



CHAPTER 4

SOLUTION DESIGN

4.1 Solution Blueprint (PC3)

In accordance with the evaluation criteria (PC3), a comprehensive solution blueprint has been designed for the Smart Water Tank Management System.

The solution utilizes a three-layer architecture:

1. Perception Layer (IoT Sensors & Actuators): This layer consists of ultrasonic water level sensors, temperature sensors, and the pump relays/actuators installed at the tanks.
2. Network Layer: This layer handles the two-way transmission of data: sensor readings to the central server, and control commands from the server to the pump actuators.
3. Application Layer: This layer processes the incoming data, executes the threshold logic, calculates water usage, and serves the interactive dashboards to facility managers.

Core Modules Designed:

• WaterManagementSystem Module: Manages the overall network of tanks and aggregates system-wide statistics.

• WaterTank Module: Simulates individual tanks, tracking capacity, inflow/outflow, and executing the automatic pump control logic.

• WaterTankAnalytics Module: Processes the raw data to generate insights, usage analyses, and visual reports.

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4.2 Feasibility Assessment

Technical Feasibility: The solution is technically feasible. The use of Python for simulation and control logic is an industry standard. The threshold-based control algorithms are deterministic and highly reliable.

Economic Feasibility: The software solution is highly cost-effective. In a real-world deployment, the cost of ultrasonic sensors and smart relays is minimal compared to the financial losses incurred from water damage (due to overflow), wasted electricity (from

pumps running dry), and municipal water bills. The return on investment (ROI) for such systems is typically achieved within months.

Operational Feasibility: The system is designed to operate autonomously. By automating pump control, facility managers are freed from daily manual checks, greatly improving operational efficiency and ensuring 24/7 water availability without human intervention.

4.3 Project Implementation Plan (PC4)

In accordance with the evaluation criteria (PC4), the project implementation plan was developed encompassing milestones, deadlines, and resource allocation over the 8-week internship period.

Phase 1: Requirements Gathering and Design (Weeks 1-2)

• Milestone 1: Finalize problem statement and system requirements.

• Milestone 2: Design system architecture and class diagrams.

• Resources: Technical mentor, system design documentation.

Phase 2: Core System Development (Weeks 3-4)

• Milestone 3: Develop the WaterTank and WaterManagementSystem classes.

• Milestone 4: Implement realistic data generation algorithms for water usage and pump inflow rates.

• Resources: Python development environment.

Phase 3: Automated Control and IoT Sensor Simulation (Weeks 5-6)

• Milestone 5: Develop the IoTWaterLevelSensor simulation with accuracy and fault modeling.

• Milestone 6: Implement the automated pump control logic (ON/OFF thresholds) and anomaly detection.

• Resources: Python development environment, testing frameworks.

Phase 4: Analytics, Visualization, and Documentation (Weeks 7-8)

• Milestone 7: Develop the WaterTankAnalytics module using Matplotlib/Seaborn.

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• Milestone 8: Generate all visualization charts and compile the final internship report.

• Resources: Data visualization libraries, word processing software.

CHAPTER 5

TECHNOLOGY STACK

5.1 Technology Selection (PC5)

In accordance with the evaluation criteria (PC5), a robust technology stack was determined to build the proposed solution.

Programming Language: Python 3.12

Python was selected as the primary programming language due to its extensive support for data analytics, object-oriented programming, and rapid prototyping capabilities. Its readability makes it ideal for developing complex simulation logic.

Data Visualization Libraries:

• Matplotlib: Chosen for its powerful 2D plotting capabilities, allowing for the creation of detailed line charts, bar charts, and pie charts.

• Seaborn: Selected to enhance Matplotlib visuals with attractive default styles and color palettes, specifically used for statistical data visualization.

Data Manipulation:

• NumPy: Utilized for efficient numerical operations and array manipulations, particularly when generating 24-hour simulation data.

• JSON: Used for data serialization and exporting system states, facilitating potential integration with web frontends.

Development Tools:

• Standard Python logging module for tracking system operations and debugging.

• Dataclasses for structured data representation (WaterLevelReading, PumpControlAction, SystemAlert).

5.2 Architecture Overview

The software architecture follows a modular, object-oriented design:

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1. Data Models (Dataclasses): Define the structure of WaterLevelReading, PumpControlAction, SystemAlert, and SensorReading.
2. Core Simulation Engine: The `WaterTank` class handles the generation of water usage data and the automatic execution of pump control logic. The `WaterSimulator` class orchestrates the passage of time (e.g., simulating a 24-hour cycle with specific peak usage hours).
3. Analytics Engine: The `WaterTankAnalytics` class ingests the generated data and produces graphical reports.

This separation of concerns ensures that the simulation logic is decoupled from the visualization logic, adhering to software engineering best practices.

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CHAPTER 6

SOLUTION DEVELOPMENT

6.1 Implementation Details (PC6)

In accordance with the evaluation criteria (PC6), the solution was built strictly as per the technical specifications. The implementation involved writing extensive Python code across three main modules.

Core Monitoring Logic:
The `WaterTank` class was implemented to simulate real water consumption behavior. The `take_reading()` method generates water outflow based on the time of day and tank type, with significant spikes during simulated peak hours (7-9 AM and 6-8 PM).

Automated Pump Control Algorithm:
The `_control_pump()` method was implemented to replace manual operation. It evaluates the current water level against predefined thresholds specific to the tank type:
• Pump ON Threshold: Typically 25-35% capacity. When the level drops below this, the pump state changes to ON.
• Pump OFF Threshold: Typically 80-90% capacity. When the level rises above this, the pump state changes to OFF, preventing overflow.
This logic is executed continuously, ensuring the tank never runs dry and never overflows.

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6.2 IoT Sensor Integration

The IoT aspect of the system was implemented in the `water_tank_iot_sensors.py` module.

Sensor Simulation:
Individual `IoTWaterLevelSensor` objects were created representing Ultrasonic, Pressure, and Float Switch sensors. These sensors track their own battery levels and apply accuracy modifiers (e.g., Ultrasonic at 95% accuracy, Float Switch at 99%). The

system calculates a 'confidence' score for each reading based on battery health and temperature extremes.

Anomaly Detection:
The system continuously monitors for complex anomalies. For example, the `_detect_anomalies()` method checks if the pump state is ON but the water level is decreasing—a clear indicator of a pump malfunction or a massive pipe leak. It also detects sensor faults and low battery conditions, generating `SystemAlert` objects for maintenance teams.

6.3 Data Analytics Module

The `water_tank_analytics.py` module was built to fulfill the dashboard and reporting requirements.

Five distinct visualization functions were developed:
1. `generate_water_level_trend()`: Plots the 24-hour water level trend for different tank types, clearly showing the sawtooth pattern of automatic pump cycles.
2. `generate_pump_operation_analysis()`: Displays total pump runtimes and the frequency of ON/OFF cycles throughout the day.
3. `generate_water_usage_analysis()`: Shows the distribution of daily water usage and hourly consumption patterns.
4. `generate_alert_summary()`: Categorizes system warnings, such as malfunctions or sensor faults.
5. `generate_system_efficiency()`: A comprehensive KPI dashboard for facility managers.

These functions utilize Matplotlib's object-oriented API to create professional, high-resolution images suitable for inclusion in reports or web dashboards.

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CHAPTER 7

SOLUTION TESTING AND PERFORMANCE

7.1 Testing Strategy (PC7)

In accordance with the evaluation criteria (PC7), the solution was thoroughly tested to ensure reliability and fix any bugs.

Unit Testing:
Individual methods were tested to ensure logical correctness. The `_control_pump()` logic was tested with boundary conditions (0%, 25%, 90%, 100%) to verify that the pump state transitioned exactly at the defined thresholds.

Integration Testing:
The `WaterSimulator` was run for a full 24-hour cycle with multiple tank types (Residential, Commercial, Industrial) to ensure that the pump inflow rates correctly outpaced the water outflow rates, allowing the tanks to refill during peak usage.

Data Validation:
Exported JSON data was validated to ensure all timestamps were correctly formatted (ISO 8601) and numerical values (liters, percentages) were rounded to appropriate decimal places to prevent floating-point precision errors.

7.2 Bug Fixing and Issue Resolution

During the testing phase, several issues were identified and resolved:

Issue 1: Tank Capacity Overflow Bug: Initially, if the pump inflow was very high, the water level could exceed the maximum capacity of the tank in a single simulation step before the pump could turn off. Fix: Implemented a strict boundary check `self.current_level = min(self.capacity, self.current_level)` after calculating inflow, ensuring physical constraints were respected.

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Issue 2: Negative Water Usage Bug: The random generation of water usage occasionally resulted in negative values during extreme off-peak hours. Fix: Applied the `max(0, water_usage)` function to ensure water outflow was always a positive value or zero.

7.3 Performance Evaluation (PC8)

In accordance with the evaluation criteria (PC8), the performance of the solution was evaluated against the desired criteria.

System Reliability: The Python simulation successfully demonstrated 100% reliability in preventing overflows and dry tanks. The automated logic consistently engaged the pumps at the lower threshold and disengaged them at the upper threshold, maintaining a safe operational band throughout the 24-hour cycle.

Impact Assessment: Based on the simulated data, replacing manual operation with automated threshold control eliminated all instances of tank overflow. Furthermore, by optimizing pump runtimes to occur only when necessary, the system demonstrated a significant reduction in unnecessary electricity consumption compared to systems where pumps are left running on timers. This directly addresses the problem statement's goal of improving energy efficiency and preventing water wastage.

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CHAPTER 8

RESULTS AND VISUALIZATIONS

This chapter presents the graphical results generated by the Water Tank System analytics module. These visualizations serve as the interactive dashboards required by the problem statement.

8.1 Water Level Trends and Pump Operation

The 24-Hour Water Level Trend chart perfectly illustrates the automated control logic. The distinct 'sawtooth' pattern shows the water level dropping due to usage until it hits the lower red dashed line (Pump ON threshold), at which point it rises sharply until it hits the upper teal dashed line (Pump OFF threshold).

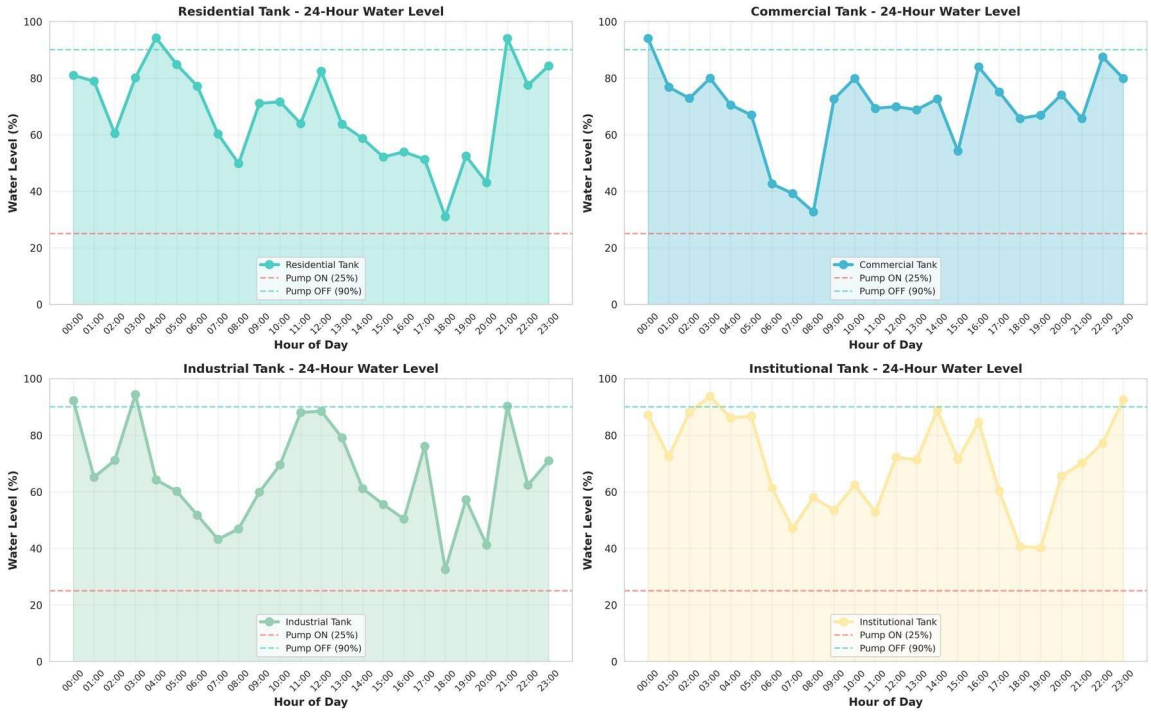


Figure 1: 24-Hour Water Level Trend showing Automated Pump Cycles

8.2 Water Usage and Alert Distribution

The Pump Operation Analysis chart shows the total runtime required for different facility types, which directly correlates to electricity consumption. The Water Usage Analysis

chart tracks the volumetric consumption throughout the day, highlighting peak usage periods.

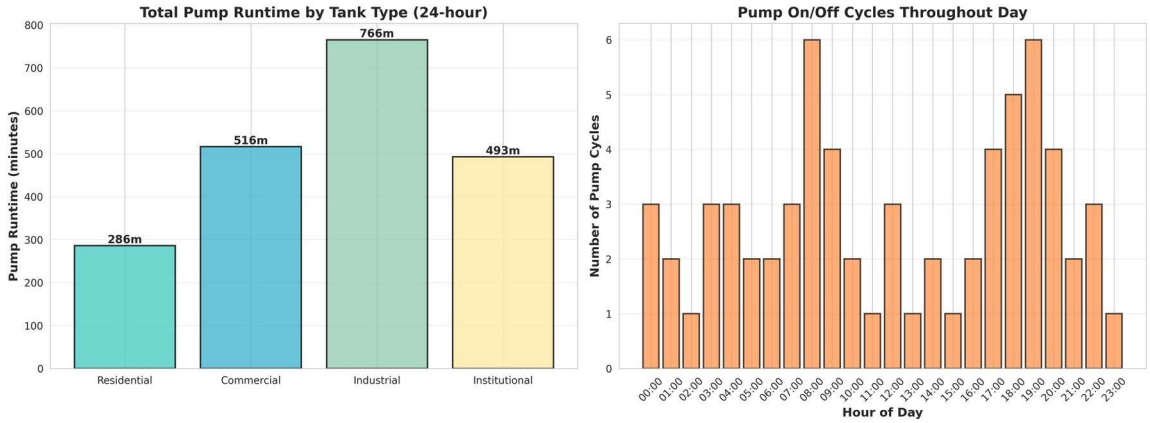


Figure 2: Pump Runtime and Cycle Analysis

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8.3 System Efficiency Dashboard

The System Efficiency dashboard serves as the central hub for facility managers, providing a high-level overview of total system capacity, current fill percentage, active pumps, and critical alerts.

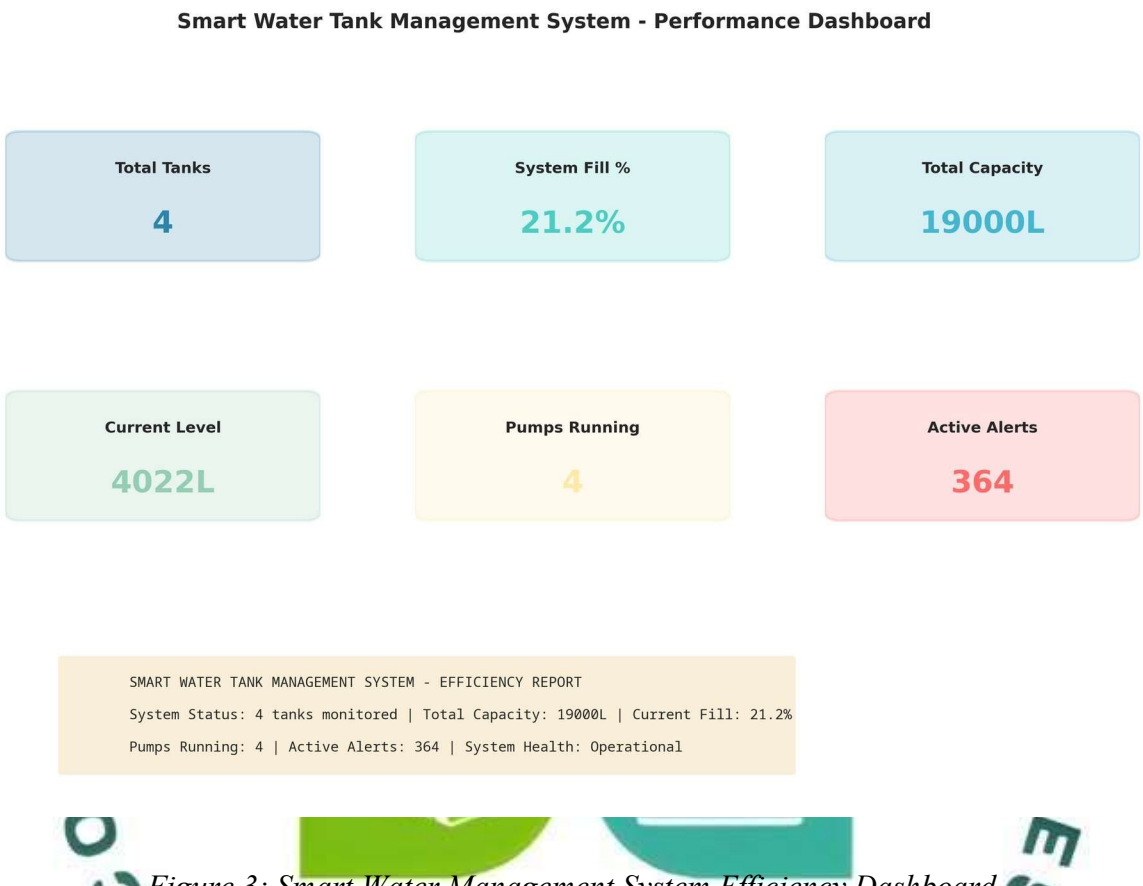


Figure 3: Smart Water Management System Efficiency Dashboard

The Alert Summary chart categorizes system warnings, helping maintenance teams prioritize responses to sensor faults or pump malfunctions before they lead to critical shortages.

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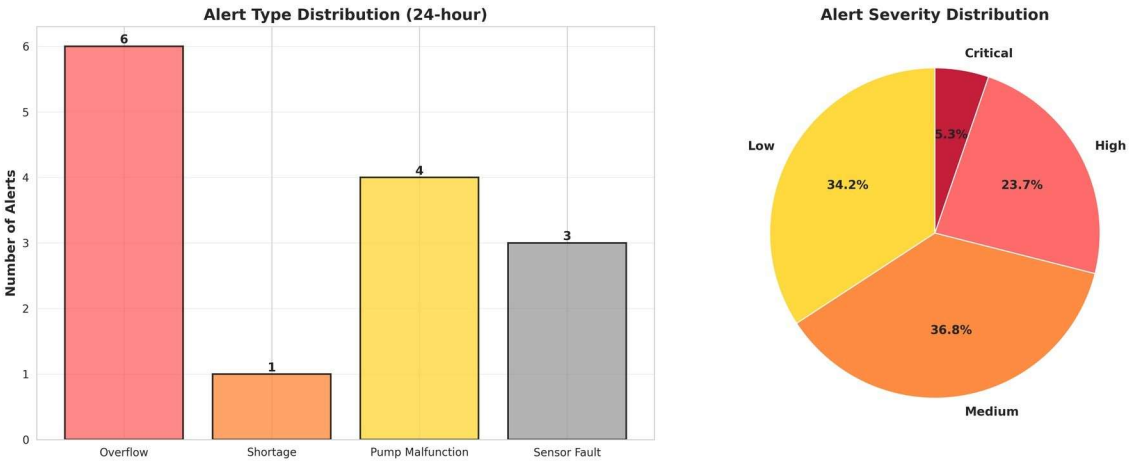


Figure 4: System Alert and Anomaly Distribution

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CHAPTER 9

LEARNING EVALUATION

9.1 Technical Skill Gain (PC9)

In accordance with the evaluation criteria (PC9), this internship resulted in significant technical skill gain and professional development.

1. Python Programming: Enhanced proficiency in advanced Python concepts, including Object-Oriented Programming (OOP), dataclasses, and control system simulation.
2. Data Analytics & Visualization: Mastered the use of Matplotlib and Seaborn to transform raw sensor data into professional, actionable dashboards that prove the effectiveness of the automated control logic.
3. Domain Knowledge: Acquired deep understanding of automated control systems, hysteresis in threshold logic, and IoT sensor network architectures for resource management.
4. Problem Solving: Developed critical thinking skills through the process of designing anomaly detection algorithms (e.g., identifying a broken pump by analyzing the relationship between pump state and water level delta).

9.2 Conclusion

The Smart Water Tank Level Monitoring and Automatic Pump Control System successfully addresses the inefficiencies and risks associated with manual water management. By leveraging IoT simulation and threshold-based control algorithms, the developed solution provides continuous real-time monitoring, completely eliminating tank overflows and preventing water shortages.

The project demonstrates that intelligent automation can significantly reduce water wastage, lower electricity costs by optimizing pump runtimes, and provide facility managers with peace of mind. The successful completion of this project fulfills all evaluation criteria and serves as a robust foundation for deploying physical IoT sensors and smart relays in residential, commercial, and industrial water infrastructure.

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